

Dated: _____

Qno. 4.

DECLARE q, p, s, a, FB : Real

INPUT q

INPUT p

INPUT d

INPUT t

IF $q \leq 0$ OR $p \leq 0$ OR $d < 0$ OR $t < 0$

THEN

PRINT ("Error")

ENDIF

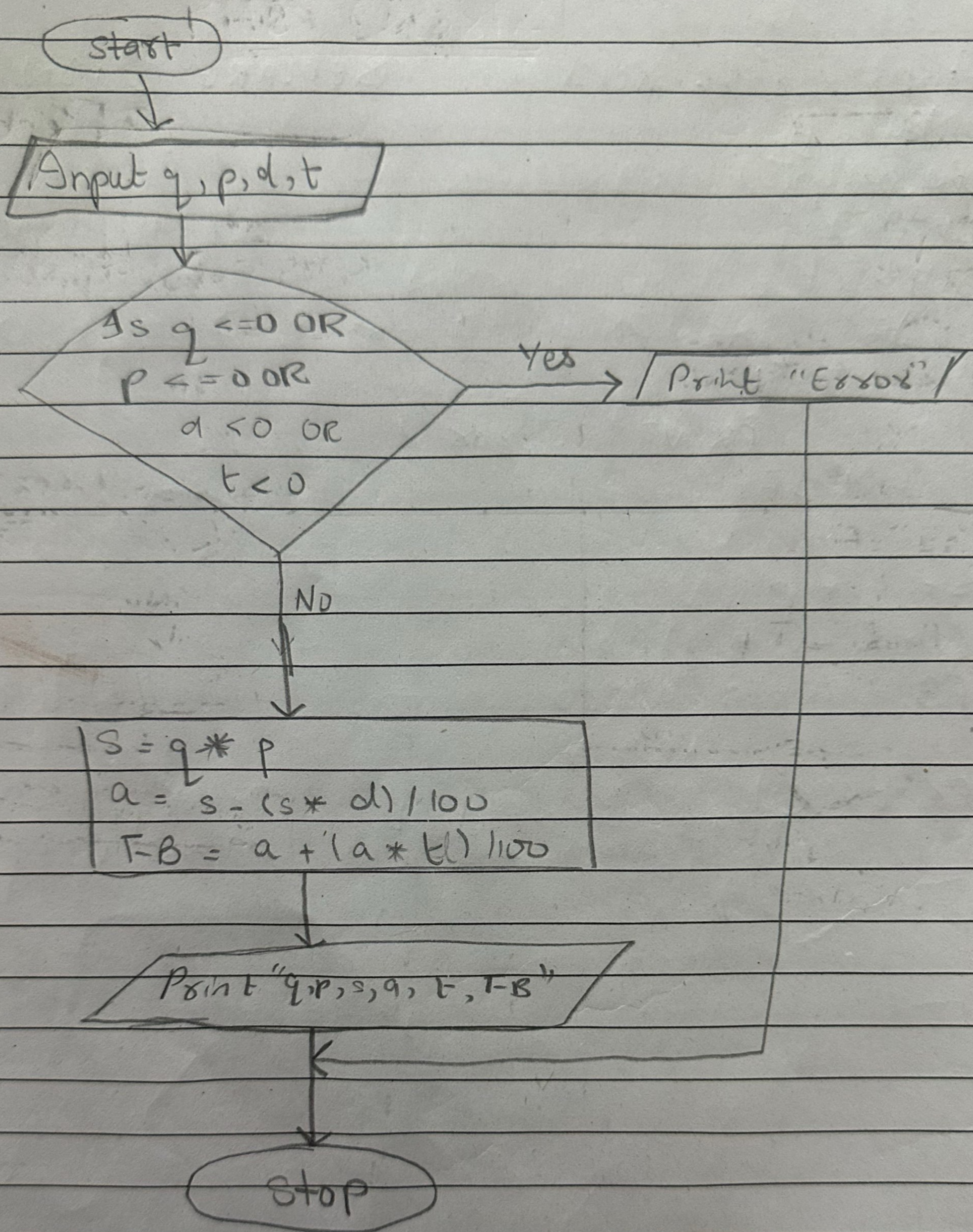
$s = q * p$

$a = s - (s * d) / 100$

$FB = a + (a * t) / 100$

// Find Bill

Print ("Bill" " " "Quantity:" q " " "Price:" p " " "Total" s
" " "Discounted amount:" a " " "Tax:" t " " "Final Bill:"
 FB)



Dated: _____

Q no. 4.

PAC chart.

INPUT

- Quantity purchased (q)
- Unit price per item (p)
- discount percentage (d)
- Tax percentage (t)

OUTPUT

- Error message if validation fails ("Error")
- Bill details
 - quantity (q)
 - price (p)
 - total (s)
 - discounted amount (a)
 - tax percentage (t)
 - Final bill (FB)

PROCESSING REQUIRED

- Read q, p, d and t
- Validate inputs: checking
 $q \leq 0$ OR $p \leq 0$ OR $d \leq 0$
OR $t \leq 0$ OR $d > 100$
 - If invalid, print error message
- Calculate total and Final Bill.

$$\text{total : } S = q * p$$

$$\text{discounted amount : } a =$$

$$S - (S * d) / 100$$

$$\text{Final Bill : } FB = a + (a * t) / 100$$

SOLUTION ALTERNATIVE.

- Upper bound for discount percentage
($d \leq 100$) can be included in the validation checks
- Can be separated into three modular functions.

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GPD Chart:

INPUT	PROCESS	OUTPUT
• q (quantity) • p (Price per item) • d (discount %) • t (tax %)	1) Input q, p, d, t 2) Check validation if $q \leq 0$ or $p \leq 0$ or $d < 0$ or $t < 0$ or $d > 100$ Print "Error" 3) Calculate total: $s = q * p$ 4) Calculate discounted amount: $a = s - (s * d) / 100$ 5) Calculate final bill $FB = a + (a * t) / 100$ 6) Print Bill (q, p, s, a, t, FB)	• Error message (if invalid) • Bill details - q, p, s, a, t, FB.